

11, 15

11 → binary form

$$\begin{array}{r} \text{step 1: } 2 \overline{) 11} \quad (5 \\ \underline{10} \\ 1 \end{array}$$

$$\begin{array}{r} \text{step 2: } 2 \overline{) 5} \quad (2 \\ \underline{4} \\ 1 \end{array}$$

$$\begin{array}{r} \text{step 3: } 2 \overline{) 2} \quad (1 \\ \underline{2} \\ 0 \end{array}$$

$$\begin{array}{r} \text{step 4: } 2 \overline{) 1} \quad (0 \\ \underline{0} \\ 1 \end{array}$$

11 → 1 0 1 1

15 → binary conversion

$$\begin{array}{r} \text{step 1: } 2 \overline{) 15} \quad (7 \\ \underline{14} \\ 1 \end{array}$$

$$\begin{array}{r} \text{step 2: } 2 \overline{) 7} \quad (3 \\ \underline{6} \\ 1 \end{array}$$

$$\begin{array}{r} \text{step 3: } 2 \overline{) 3} \quad (1 \\ \underline{2} \\ 1 \end{array}$$

$$\begin{array}{r} \text{step 4: } 2 \overline{) 1} \quad (0 \\ \underline{0} \\ 1 \end{array}$$

15 → 1 1 1 1

Bitwise AND operator

11	15	11 & 15
1	1	1
0	1	0
1	1	1
1	1	1

$$= 2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 1$$

$$= 8 + 0 + 2 + 1$$

$$= 11$$

$$11 \& 15 = 11$$

Bitwise OR operator

11	15	11 15
1	1	1
0	1	1
1	1	1
1	1	1

$$= 2^3 \times 1 + 2^2 \times 1 + 2^1 \times 1 + 2^0 \times 1$$

$$= 8 + 4 + 2 + 1$$

$$= 15$$

$$11 | 15 = 15$$

Bitwise XOR operator

11	15	11 ^ 15
1	1	0
0	1	1
1	1	0
1	1	0

$$= 2^3 \times 0 + 2^2 \times 1 + 2^1 \times 0 + 2^0 \times 1$$

$$= 0 + 4 + 0 + 0$$

$$= 4$$

$$11 ^ 15 = 4$$



12, 13

12 $\rightarrow$ binary form $\rightarrow 1100$

13 $\rightarrow$ binary form $\rightarrow 1101$

Bitwise and operator

Bitwise OR operator

Bitwise XOR operator

Step 1:
$$\begin{array}{r} 2 \overline{) 12} \quad (6 \\ \underline{12} \\ 0 \end{array}$$

Step 2:
$$\begin{array}{r} 2 \overline{) 6} \quad (3 \\ \underline{6} \\ 0 \end{array}$$

Step 3:
$$\begin{array}{r} 2 \overline{) 3} \quad (1 \\ \underline{2} \\ 1 \end{array}$$

Step 4:
$$\begin{array}{r} 2 \overline{) 1} \quad (0 \\ \underline{0} \\ 1 \end{array}$$

Step 2:
$$\begin{array}{r} 2 \overline{) 13} \quad (6 \\ \underline{12} \\ 1 \end{array}$$

Step:
$$\begin{array}{r} 2 \overline{) 6} \quad (3 \\ \underline{6} \\ 0 \end{array}$$

Step 3:
$$\begin{array}{r} 2 \overline{) 3} \quad (1 \\ \underline{2} \\ 1 \end{array}$$

Step 4:
$$\begin{array}{r} 2 \overline{) 1} \quad (0 \\ \underline{0} \\ 1 \end{array}$$

Bitwise AND operator

Bitwise OR operator

Bitwise XOR operator

12	13	12 & 13
1	1	1
1	1	1
0	0	0
0	1	0

12	13	12 13
1	1	1
1	1	1
0	0	0
0	1	1

12	13	12 ^ 13
1	1	0
1	1	0
0	0	0
1	1	0

$$= 2^3 \times 1 + 2^2 \times 1 + 2^1 \times 0$$

$$+ 2^0 \times 0$$

$$= 8 + 4 + 0 + 0$$

$$= 12$$

$$= 2^3 \times 1 + 2^2 \times 1 +$$

$$2^1 \times 0 + 2^0 \times 1$$

$$= 8 + 4 + 0 + 1$$

$$= 13$$

$$= 2^3 \times 0 + 2^2 \times 0 + 2^1 \times 0$$

$$+ 2^0 \times 1$$

$$= 0 + 0 + 0 + 0$$

$$= 0$$



10 & 14

Bitwise & operator

10 → binary form → 1010

14 → binary form → 1110

10	14	10 & 14
1	1	1
0	1	0
1	1	1
0	0	0

$$= 2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 0$$

$$= 8 + 0 + 2 + 0$$

$$= 10$$

Bitwise OR operator

10	14	10 14
1	1	1
0	1	1
1	1	1
0	0	0

$$= 2^3 \times 1 + 2^2 \times 1 + 2^1 \times 1 + 2^0 \times 0$$

$$= 8 + 4 + 2 + 0$$

$$= 14$$

XOR operator

10	14	10 ^ 14
1	1	0
0	1	1
1	1	0
0	0	0

$$= 2^3 \times 0 + 2^2 \times 1$$

$$+ 2^1 \times 0 + 2^0 \times 0$$

$$= 4$$